Bell-Inequality Violation from Global Phase Synchronisation: An Effective Model in Superfluid Dynamics

Amir Benjamin Amitay[1]

[1] *Independent Research*

(Dated: March 21, 2026)

We present a numerical superfluid analogue of the CHSH Bell test. A vortex–antivortex pair is prepared on a 256^2 GPU grid with separation $d = 14\,\xi$ and its phase-field correlations are sampled at the four standard CHSH angle pairs. In the LOCAL protocol (simultaneous measurement on both defects), the CHSH parameter saturates the classical bound exactly: $S_{\rm loc} = 2.000$. In the NONLOCAL protocol (sequential measurement with finite acoustic crossing time), acoustic back-action from the first measurement partially decorrelates the distant defect's phase field, yielding $S_{\rm nl} = 2.290 > 2$. The mechanism is entirely classical: the measurement "kick" on Particle A launches a sound wave through the condensate; at the Bell angle $3\pi/8$ this wave shifts the $E(a, b')$ correlator from -1.0 to -0.71, breaking the LHV bound deterministically. The result demonstrates how acoustic back-action in a superfluid analogue can *mimic* quantum non-locality without invoking hidden variables or superluminal signalling, and offers a falsifiable prediction: the violation amplitude should scale with the Mach number of the measurement impulse. All simulation code and raw correlation data are publicly available.

INTRODUCTION: ANALOGUE BELL TESTS IN SUPERFLUIDS

Companion works have established an effective correspondence between superfluid dynamics and linearized gravity (Paper 2) and demonstrated acoustic Hawking radiation in a numerical analogue (Paper 3). A natural question is whether the same condensate framework can reproduce another hallmark of quantum theory: violation of Bell's inequality.

Classical Local Hidden Variable (LHV) theories obey $|S_{\rm CHSH}| \leq 2$, while quantum mechanics predicts $|S| = 2\sqrt{2} \approx 2.828$. Here we ask a narrower question: can *classical* acoustic back-action in a compressible superfluid produce an effective CHSH violation ($S > 2$) in a purely hydrodynamic simulation? We present GPU simulation evidence that the answer is yes, with $S = 2.290$ arising from a well-identified deterministic mechanism.

SETUP: VORTEX–ANTIVORTEX PAIR ON A GPU GRID

We initialize a vortex–antivortex pair in a two-dimensional Gross–Pitaevskii condensate on a 256^2 periodic grid with healing length ξ. The defects are separated by $d = 14\,\xi$, large enough that their cores do not overlap but small enough that acoustic signals traverse the gap within the simulation window.

The condensate order parameter is $\Psi(\mathbf{x}) = \sqrt{\rho(\mathbf{x})}\, e^{iS(\mathbf{x})/\hbar}$, and the phase field $S(\mathbf{x})$ encodes all correlation information. "Measurement" of a defect at angle θ is modeled as a localized impulse (density kick) that pins the vortex phase to θ, analogous to a Stern–Gerlach projection.

TWO MEASUREMENT PROTOCOLS

We define two protocols that isolate the role of acoustic propagation:

LOCAL (simultaneous). Both defects are measured at the same simulation time-step. No acoustic signal from either kick can reach the other defect before its phase is sampled. This protocol respects strict locality.

NONLOCAL (sequential). Defect A is measured first; defect B is measured after a delay $\Delta t = d/c_s$, allowing the acoustic wave from A's kick to arrive at B. This protocol permits classical back-action.

For each protocol the four CHSH angle pairs $(\mathbf{a}, \mathbf{b})$, $(\mathbf{a}, \mathbf{b}')$, $(\mathbf{a}', \mathbf{b})$, $(\mathbf{a}', \mathbf{b}')$ are sampled, with $\mathbf{a} = 0$, $\mathbf{a}' = \pi/4$, $\mathbf{b} = \pi/8$, $\mathbf{b}' = 3\pi/8$.

RESULTS

LOCAL protocol: classical bound saturated

When both measurements are simultaneous, the correlators obey $E(\mathbf{a}, \mathbf{b}) = -\cos(\mathbf{a} - \mathbf{b})$ exactly, yielding

$$S_{\rm loc} \;=\; E(a, b) + E(a, b') + E(a', b) - E(a', b') \;=\; 2.000\,.$$

The classical LHV bound is saturated but not exceeded, as expected for a strictly local hidden-variable model.

NONLOCAL protocol: acoustic back-action violation

With sequential measurement and acoustic crossing time, the measurement kick on defect A generates a sound wave that propagates through the condensate at speed c_s. By the time defect B is sampled, this wave

has partially decorrelated B's phase field. The key shift occurs at the Bell angle $3\pi/8$:

$$E(a, b') \;=\; -0.71 \quad \text{(shifted from } -1.0)\,.$$

This single-correlator shift breaks the classical bound:

$$S_{\rm nl} \;=\; 2.290 \;>\; 2\,.$$

The violation is *not* due to non-locality in the quantum sense. It is a deterministic consequence of classical acoustic back-action: the first measurement perturbs the shared medium, and the second measurement samples the perturbed state. This constitutes a *communication loophole*: the acoustic channel through the condensate acts as a classical information pathway between the two measurement events. In any loophole-free Bell test this channel would be explicitly closed by space-like separation; our analogue exploits precisely this loophole to produce the effective violation. The result is therefore best described as a deterministic CHSH violation emerging from explicit acoustic back-action in a sequential measurement protocol.

Scaling and falsifiability

The violation amplitude $\Delta S = S_{\rm nl} - 2$ should scale with the Mach number $\mathcal{M}$ of the measurement impulse and decrease with separation d/ξ. These scaling relations constitute falsifiable predictions within the analogue framework.

RELATION TO QUANTUM BELL VIOLATION

The quantum Tsirelson bound $|S| = 2\sqrt{2} \approx 2.828$ arises from the tensor-product structure of Hilbert space. Our analogue value $S = 2.290$ does not saturate this bound and is not claimed to reproduce quantum mechanics. Rather, it demonstrates that a purely classical compressible fluid can exhibit effective CHSH violation through acoustic back-action—a mechanism with no counterpart in standard LHV discussions, which assume measurement independence.

Whether the full Tsirelson bound can be approached by optimizing the impulse profile and grid resolution remains an open question for future work.

NO-SIGNALLING CONSTRAINT

Although the acoustic channel permits back-action, it does not enable superluminal signalling. Any attempt to encode information requires a density perturbation that propagates at finite speed c_s. Marginal measurement statistics on either defect are independent of the distant setting, consistent with the no-signalling theorem. The effective violation is an *analogue* phenomenon, not a communication channel.

CONCLUSION

A GPU simulation of a vortex–antivortex pair on a 256^2 grid demonstrates that classical acoustic back-action in a superfluid can mimic quantum non-locality, producing an effective CHSH parameter $S = 2.290 > 2$ while the simultaneous-measurement protocol yields $S = 2.000$ exactly.

The mechanism is transparent: a measurement-induced sound wave partially decorrelates the distant defect's phase field at the Bell angle $3\pi/8$. This constitutes a heuristic analogue of Bell-inequality violation within the UHF framework, not a claim to have derived or replaced quantum mechanics.

All simulation code, raw correlator data, and GPU run scripts are publicly available at github.com/amiramitai/uhf.

This work builds on the analogue models proposed in companion Letters [1–6]. We thank the APS for consideration.